

Inferential Privacy, Impossibility, and Differential Privacy

Based on Cynthia Dwork

Motivation

- ▶ A statistical database is meant to reveal useful information about a population or dataset.
- ▶ At the same time, it should protect the privacy of the individuals represented in it.
- ▶ This creates a central tension:

utility vs. privacy.

- ▶ A rigorous treatment requires clear definitions of:
 - ▶ what counts as a privacy breach,
 - ▶ what auxiliary information the adversary may possess,
 - ▶ and what useful information the mechanism is intended to reveal.

Dalenius' Goal

Dalenius-style desideratum

Nothing about an individual should be learnable from the database that could not be learned without access to the database.

- ▶ This resembles the later cryptographic notion of *semantic security*.
- ▶ The adversary may already possess substantial outside information.
- ▶ The hope is that access to the database should not increase what can be inferred about an individual.

The Core Problem

- ▶ The database may release only aggregate statistics.
- ▶ Yet these statistics can interact with outside knowledge in dangerous ways.
- ▶ Thus privacy loss is often caused not by the release alone, but by:
 - ▶ released information, together with
 - ▶ auxiliary information already available to the adversary.

Key idea

Even a seemingly harmless statistic may enable a sensitive inference when combined with side information.

Example: Katrina Kaif's Height

Suppose:

- ▶ the database releases the average height of Indian women;
- ▶ an adversary knows that KK is exactly 3 inches taller than the average Indian woman.

Then:

- ▶ with access to the database output, the adversary can infer KK's exact height;
- ▶ without access to the database, the adversary cannot.

Lesson

The database output may enable an inference even though it does not directly mention the individual.

Why This is Striking

The paper emphasizes two remarkable features of this phenomenon:

- ▶ the privacy breach can occur even if KK is *not* in the database;
- ▶ the Dalenius-style goal cannot, in general, be achieved at all.

Consequence

Absolute inferential privacy is too strong a goal once arbitrary auxiliary information is allowed.

Formal Setting

We consider:

- ▶ a database DB drawn from a distribution D ;
- ▶ a sanitization mechanism $San(DB)$;
- ▶ a privacy breach predicate, represented by a machine C ;
- ▶ an auxiliary information generator $X(D, DB)$ producing side information z .

Two entities are compared:

- ▶ an adversary A that sees both $San(DB)$ and z ;
- ▶ a simulator A^* that sees only z .

The question is whether access to the sanitized database gives a significant advantage.

Impossibility Theorem: Informal Statement

Informal theorem

For any sanitization mechanism $San()$ and privacy breach notion C , there exists auxiliary information such that the adversary can succeed substantially better with access to the sanitized output than any simulator can succeed with the auxiliary information alone.

Symbolically:

$$\Pr[A \text{ wins}] - \Pr[A^* \text{ wins}] \geq \Delta$$

for a suitably large constant Δ .

Idea of the Proof

Very roughly, the proof works as follows:

- ▶ the database is assumed to reveal some useful information, encoded as a utility vector w ;
- ▶ the auxiliary information generator hides a privacy breach y using randomness extracted from w ;
- ▶ the adversary, who learns enough about w from the database output, can recover that hidden breach;
- ▶ the simulator, who sees only the auxiliary string, cannot.

Intuition

The released utility itself becomes a source of shared information that the adversary can exploit.

Utility is the Source of the Problem

Why does the analogy with semantic security fail?

- ▶ In cryptography, a ciphertext is designed to reveal nothing to an eavesdropper.
- ▶ In a statistical database, the released output is designed to be useful.
- ▶ Therefore the user/adversary necessarily learns something meaningful from it.

Tension

Utility makes absolute semantic-style privacy impossible.

Anonymization as a Privacy Strategy: Narayanan & Shmatikov (2008)

- ▶ Common approach:
 - ▶ Remove identifiers (name, address, etc.)
 - ▶ Release dataset for analysis
- ▶ Assumption:
 - ▶ Anonymized data protects individuals

Question

Is anonymization sufficient to prevent inference about individuals?

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- ▶ Dataset released for recommendation research
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- ▶ $\sim 500,000$ users, millions of ratings
- ▶ Identifiers removed
- ▶ Each user represented by:
 - ▶ movie ratings
 - ▶ timestamps

Key Property

High-dimensional, sparse behavioral data

De-anonymization Result

- ▶ Adversary knows:
 - ▶ a few movie ratings of a target user
 - ▶ approximate dates
- ▶ Auxiliary dataset:
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Result

A small amount of auxiliary information suffices to uniquely identify users

Mechanism of Attack

- ▶ Each user has a **rating vector**
- ▶ Vectors are:
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 - ▶ highly distinctive
- ▶ Matching procedure:
 - ▶ compare known ratings with anonymized records
 - ▶ find best similarity match

Key Insight

Human behavior is highly identifying

Robustness of the Attack

- ▶ Works even with:
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Conclusion

Anonymization + noise is insufficient in high-dimensional data

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- ▶ Entire rating history
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Important

Inference is enabled by combining:

- ▶ released data
- ▶ auxiliary information

Other Examples of De-anonymization

- ▶ **AOL Search Logs (2006)**
 - ▶ Released “anonymized” search queries
 - ▶ Users identified via unique query patterns (e.g., location, interests)
- ▶ **Massachusetts Medical Records (Sweeney)**
 - ▶ Voter rolls + anonymized hospital data
 - ▶ Re-identified Governor using ZIP code, birth date, gender
- ▶ **Mobility / Location Data**
 - ▶ Few spatio-temporal points uniquely identify individuals
 - ▶ Even coarse location traces are highly distinctive
- ▶ **Social Network De-anonymization**
 - ▶ Graph structure used as fingerprint
 - ▶ Nodes matched across anonymized and public graphs
- ▶ **Genomic Data**
 - ▶ Individuals identified via genetic markers
 - ▶ Linkage attacks using public genealogy databases

Common Pattern

Released data + auxiliary information $\Rightarrow$ unique identification

Connection to Dwork's Impossibility

Dwork's theorem:

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Netflix case is a concrete instantiation:

- ▶ Utility: rich rating dataset
- ▶ Auxiliary info: IMDb ratings
- ▶ Outcome: identity + sensitive traits

Connection to Dwork's Impossibility

Dwork's theorem:

- ▶ Auxiliary information + useful output $\Rightarrow$ inference

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Interpretation

This is not an accident—it is **inevitable**

Takeaways

- ▶ De-anonymization is a manifestation of inferential privacy failure
- ▶ Auxiliary information is the central threat
- ▶ High-dimensional datasets are inherently identifying
- ▶ This motivates:
 - ▶ differential privacy
 - ▶ formal, worst-case guarantees

A Change in Perspective

Since absolute disclosure prevention is impossible, we change the privacy objective.

Instead of asking:

Can a bad inference ever occur?

we ask:

How much more likely does participation in the database make that inference?

New idea

Privacy should measure the *additional risk incurred by participating* in the dataset.

Differential Privacy

Definition

A randomized mechanism K gives ε -differential privacy if for all datasets D_1 and D_2 differing in at most one element, and for all measurable sets of outputs S ,

$$\Pr[K(D_1) \in S] \leq e^\varepsilon \Pr[K(D_2) \in S].$$

- ▶ The output distribution should not change much when one individual's data is added or removed.
- ▶ Thus participation has only a limited effect on what can be inferred from the released output.

Interpretation of Differential Privacy

Differential privacy provides a *relative* guarantee:

- ▶ a bad event may still occur;
- ▶ but it should be almost as likely even if a given individual had not participated.

Hence:

- ▶ the individual's additional privacy risk is small;
- ▶ there is little incentive to hide or falsify one's data purely for privacy reasons.

Group Privacy

Differential privacy also extends naturally to groups.

If the datasets differ in c rows rather than 1, then the probability ratio degrades to approximately

$$e^{\epsilon c}.$$

- ▶ Small groups may still receive meaningful protection.
- ▶ For large groups, the guarantee necessarily weakens.

Interpretation

Differential privacy is fundamentally tailored to controlling the effect of *small changes* in the database.

Achieving Differential Privacy

The basic mechanism is simple:

- ▶ compute the true answer $f(X)$ to a query on database X ;
- ▶ add carefully calibrated random noise.

So the released answer is:

$$K_f(X) = f(X) + \text{noise}.$$

The amount of noise depends on how much a single individual can change the query answer.

Sensitivity

Definition

For a query function $f : D \rightarrow \mathbb{R}^d$, the L_1 -sensitivity is

$$\Delta f = \max_{D_1, D_2} \|f(D_1) - f(D_2)\|_1,$$

where D_1 and D_2 differ in at most one element.

- ▶ Δf measures the maximum influence of a single participant.
- ▶ It depends only on the query, not on the particular database instance.

Example of Sensitivity

For a counting query,

“How many rows satisfy property P ?”

changing one row can change the answer by at most 1.

Therefore:

$$\Delta f \leq 1.$$

- ▶ Low sensitivity means less noise is needed.
- ▶ High sensitivity requires more noise to preserve privacy.

Noise Distribution

The paper considers adding symmetric exponential noise coordinate-wise, with density proportional to

$$\Pr[K_f(X) = a] \propto \exp\left(-\frac{\|f(X) - a\|_1}{\sigma}\right).$$

- ▶ The parameter σ controls the noise scale.
- ▶ Larger σ means more noise and stronger privacy.

In modern terminology, this is the Laplace mechanism in the L_1 setting.

Privacy Guarantee from Sensitivity

Theorem

For $f : D \rightarrow \mathbb{R}^d$, the mechanism K_f gives

$$\frac{\Delta f}{\sigma}\text{-differential privacy.}$$

Hence, to achieve ε -differential privacy, it suffices to choose noise scale so that

$$\sigma \geq \frac{\Delta f}{\varepsilon}.$$

- ▶ privacy is controlled by ε ;
- ▶ accuracy is controlled by the size of the added noise.

Histogram Queries

A useful illustration is the histogram query:

- ▶ partition the domain into k cells;
- ▶ output the vector of counts in all cells.

Although the output has k coordinates, if one individual's data changes, only one cell count changes by 1.

Therefore:

$$\Delta f = 1.$$

Important point

Sensitivity is a property of the whole query, not simply the number of coordinates in its output.

Adaptive Queries

The paper also treats adaptive interaction:

- ▶ later queries may depend on earlier released answers;
- ▶ the mechanism still preserves differential privacy if noise is calibrated to the maximum sensitivity of the strategy.

Thus differential privacy is robust not only for one-shot queries, but also for interactive analysis.

Summary

- ▶ Dalenius' inferential privacy goal is natural but unattainable in general.
- ▶ Auxiliary information makes absolute disclosure prevention impossible.
- ▶ This motivates a shift from absolute guarantees to relative guarantees.
- ▶ Differential privacy bounds how much participation in the database can change output probabilities.
- ▶ The basic method is to add noise calibrated to the sensitivity of the query.

Takeaway

Main conceptual shift

We do not try to prevent all possible inferences.
Instead, we ensure that being in the database does not substantially increase the risk of those inferences.